IES ACADEMY • ENGINEERING

GenAI for Engineers

Version v2.2 · 14 October 2026

IES Academy

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COURSE ADMINISTRATION

Welcome & Housekeeping

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ICE-BREAKER

Let's Know Each Other

1. Your name, role and one maintenance reporting task.
2. Which app holds the work-order source?
3. Which date and status definitions determine overdue work?
4. Check which Copilot buttons and skills menus are available in your assigned apps.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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Access Course Materials

1. Open the IES Academy course folder supplied by your trainer
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the labs folder for all 10 lab folders
5. Keep the files together as you work

Course materials: trainer-supplied IES Academy folder

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Lesson Plan — 1 Day, 7.5 instructional hours

Course sequence

- Topic 1: Fundamentals of Microsoft 365 Copilot (Labs 1–2)
- Topic 2: Personalisation, .Rules and Skills for Excel (Labs 3–6)
- Topic 3: Interactive Dashboards with Custom Skills (Labs 7–8)
- Topic 4: Create a Professional Presentation with PowerPoint Copilot (Labs 9–10)

Day format

- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
  - 7.5 hours · 10 connected labs
- Learning arc
  - Concept → demonstrate → build → verify → discuss → recap

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Connected Copilot workflow

Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.

2

Reusable maintenance skills

Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.

3

Interactive dashboard

Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.

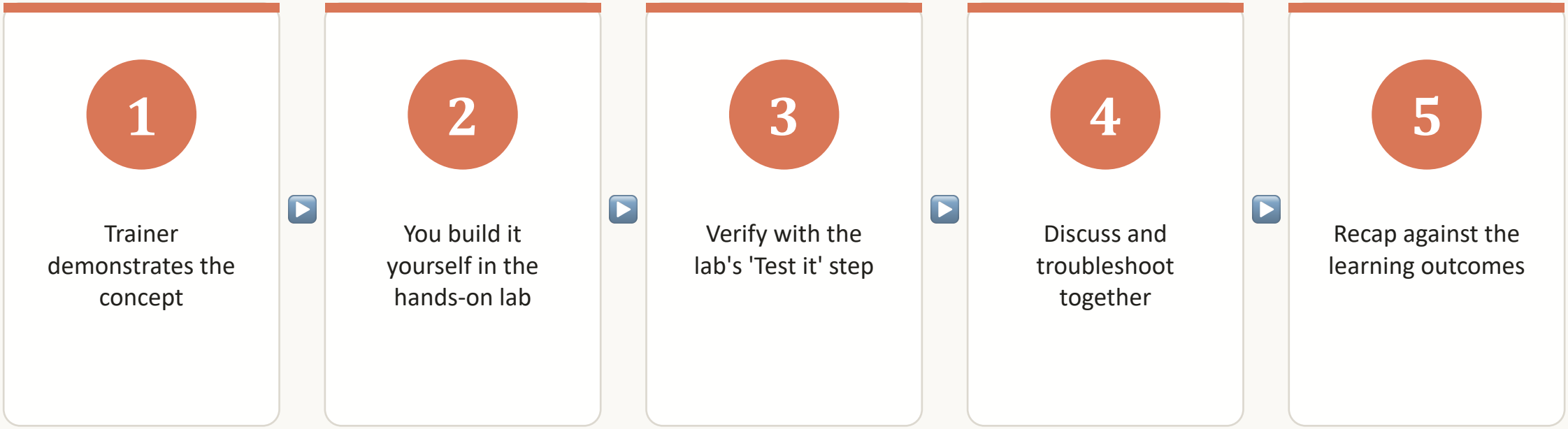
4

Professional presentation

Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.

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How You'll Learn



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TOPIC 1

Fundamentals of Microsoft 365 Copilot

One verified maintenance workflow across four apps

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Key Concepts — Fundamentals of Microsoft 365 Copilot

1

Grounding

Select actual sources in every app.

2

Work orders

IDs, status, category, site and current assignment.

3

Prompts

Goal, context, source and expectations.

4

Handoffs

Reconcile Word, Excel, slides and draft email.

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Hands-On Labs — Fundamentals of Microsoft 365 Copilot

Lab 1

- **Excel to Word**
  - Prepare Work Orders and Draft the Operations Brief

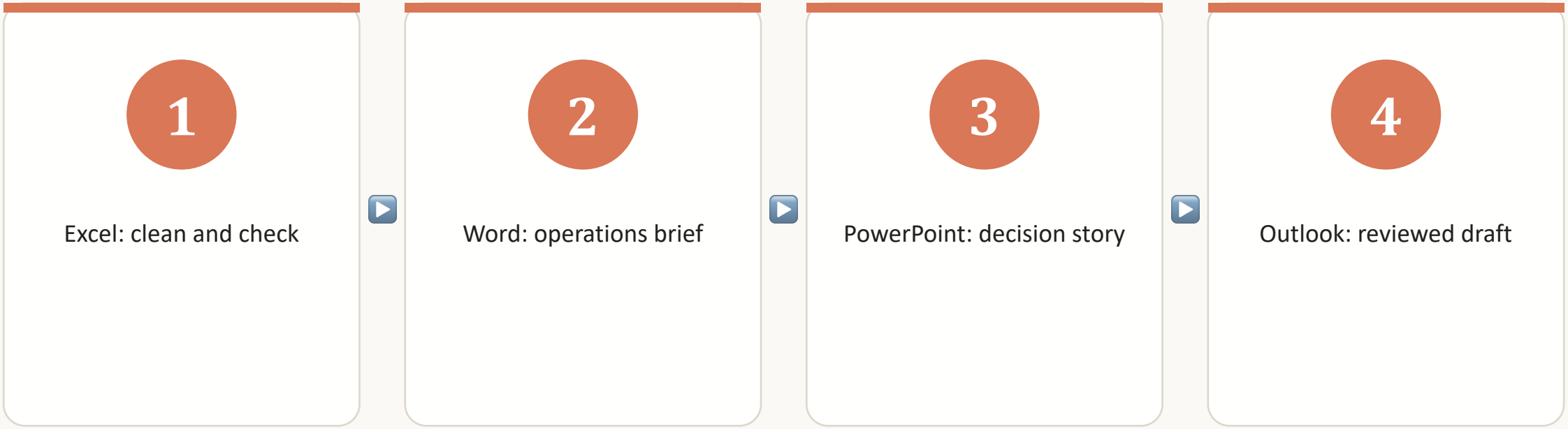
Lab 2

- **Word to PowerPoint and Outlook**
  - Create the Cross-App Maintenance Update

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CONNECTED USE CASE

The Northstar Maintenance Evidence Chain



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EDITABLE DATA MODEL

A Work Order Has State, Scope and Ownership

ID
WO-2026-0003

Site
Woodlands Hub

Category
Leak Repair

Status
In Progress

Owner
Chloe Goh

Current owner is evidence. Proposed owner is a separate planning output.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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Grounding Across Microsoft 365

WHAT IT IS

- Grounding gives an AI request the specific evidence it should use.

WHY IT MATTERS

- A maintenance figure can be correct in Excel yet become wrong in an email when the source, reporting date or filter scope changes.

CHECK IT

- Excel analyses tblWorkOrders; Word receives the checked summary.

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Grounding Across Microsoft 365

1. Identify the authoritative work-order table.
2. Provide goal, context, source and expected output.
3. Review against the named evidence.
4. Pass a verified snapshot to the next app.

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Apply the Concept — Grounding Across Microsoft 365

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- PowerPoint receives the reviewed Word brief; Outlook receives the same checked values.
- An Excel chat transcript is not an automatic PowerPoint data source.

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Work-Order Grain and Reporting Measures

WHAT IT IS

- A row represents one work order, identified by Work_Order_ID.

WHY IT MATTERS

- An exact duplicated export row changes counts and cost without representing new work.

CHECK IT

- 122 raw rows contain 120 distinct work-order IDs.

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Work-Order Grain and Reporting Measures

1. Define one ID per work order.
2. Find exact repeated export records.
3. Standardise labels without inventing facts.
4. Reconcile cleaned counts and cost.

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Apply the Concept — Work-Order Grain and Reporting Measures

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Total = Closed + Cancelled + Backlog after exact duplicate removal.
- Due on 30 September is not overdue at the 30 September snapshot.

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Prepare Work Orders and Draft the Operations Brief

LAB 1

You are Northstar Water Services' maintenance coordinator. A Q3 export contains repeated IDs and inconsistent category labels.

You'll build

Northstar_Clean_Work_Orders.xlsx and Northstar_Operations_Brief.docx, with reconciled counts, reporting scope and a source register.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive

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Sample Prompts — Lab 1

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 1

1

Clean table

120 unique Work_Order_ID rows; five canonical job categories.

2

Portfolio identity

120 = 48 Closed + 12 Cancelled + 60 backlog.

3

Evidence handoff

Overdue 50; unassigned backlog 6; total cost S$35,827.50. Word agrees with Excel.

4

No guessed owner

Blank assignments remain blank; proposals labelled unapproved.

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Prompts and Verified App Handoffs

WHAT IT IS

- A useful prompt combines the intended goal, operational context, named source and output expectations.

WHY IT MATTERS

- The same number may have different meaning when a chart counts all work orders while an email calls it open backlog.

CHECK IT

- Goal: draft a maintenance update for the operations manager.

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Prompts and Verified App Handoffs

1. Write the goal and audience.
2. Attach or select the actual source.
3. Specify definitions and missing-value behaviour.
4. Reconcile the returned claims.

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Apply the Concept — Prompts and Verified App Handoffs

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Source: reviewed Northstar operations brief, all filters All, 30 September.
- Expectation: label assignment suggestions as proposals and leave unknown approvals UNKNOWN.

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Outlook Drafting and Operational Decisions

WHAT IT IS

- Copilot can help structure a draft email from supplied context.

WHY IT MATTERS

- An apparently routine update can accidentally promise technician reassignment or imply that a manager approved a recovery plan.

CHECK IT

- Current owner blank; proposed owner Aisha remains unapproved.

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Outlook Drafting and Operational Decisions

1. Provide the synthetic thread and checked facts.
2. Draft a concise update.
3. Review recipients, facts and action wording.
4. Save the unsent draft and evidence.

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Apply the Concept — Outlook Drafting and Operational Decisions

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A draft addressed to example.invalid is a training artifact.
- The handoff says decision requested, not decision approved.

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Create the Cross-App Maintenance Update

LAB 2

Continue the verified operations brief from Lab 1. You will turn it into an initial PowerPoint update, then prepare an Outlook reply to a synthetic maintenance thread.

You'll build

Northstar_Initial_Update.pptx, an unsent Outlook draft and Northstar_Cross_App_Evidence.csv with Excel, Word, PowerPoint and Outlook figures reconciled.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive

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Sample Prompts — Lab 2

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 2

1

Four apps

Excel Checks, Word brief, four-slide update and unsent Outlook draft share the same date and values.

2

Cross-app grounding

Named file or pasted verified evidence is provided in each app; no assumed shared chat.

3

Draft boundary

example.invalid recipient; nothing sent and no assignment approved.

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TOPIC RECAP

Recap - Fundamentals of Microsoft 365 Copilot

1. LO1: Same work-order evidence across four apps.

2. Verify: 120 clean IDs and reconciled brief/update/draft.

TOPIC 2

Personalisation, .Rules and Skills for Excel

Account preferences, workbook conventions and reusable methods

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Key Concepts — Personalisation, .Rules and Skills for Excel

1

Personalisation

Account preferences across workbooks; task-specific overrides.

2

.Rules

Visible column-A instructions travel with the workbook.

3

Pareto Skill

Create, install and apply verified contribution analysis.

4

Reporting Skill

Excel and PowerPoint use with input and rerun checks.

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Hands-On Labs — Personalisation, .Rules and Skills for Excel

Labs 3–4

- **Personalisation**
  - Personalise Copilot in Excel for Consistent Service Reports
- **Workbook .Rules**
  - Create and Test Workbook .Rules for Water Service Reporting

Labs 5–6

- **Custom Pareto Chart Skill**
  - Create a Custom Pareto Chart Skill and Apply It
- **Reporting Skill**
  - Create, Use and Improve the Maintenance Reporting Skill

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Four Scopes of Customisation

1

Personalisation: your account

2

.Rules: this workbook

3

Skill: repeated method

4

Prompt: this request

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Account Personalisation and Specific Requests

WHAT IT IS

- Excel Copilot Personalisation stores account-level preferences for display, writing style and formula conventions.

WHY IT MATTERS

- Consistent S$ display and short explanations reduce repetitive requests without changing numeric values.

CHECK IT

- A value of 35827.5 can display as S$35,827.50 or S$35,827.5000.

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Account Personalisation and Specific Requests

1. Choose recurring display preferences.
2. Save them to the account.
3. Compare two workbook outputs.
4. Test a specific request and restore.

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Apply the Concept — Account Personalisation and Specific Requests

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Changing a number format does not change its numeric value.
- Another user opening the file does not receive your account preferences.

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Personalise Copilot in Excel for Consistent Service Reports

LAB 3

Configure temporary account-level display and explanation preferences, test them in two copies of the same verified Northstar workbook, and distinguish preferences from workbook-specific definitions. A task-specific prompt can supersede Personalisation.

You'll build

Northstar_Personalisation_Test.xlsx, a two-workbook preference comparison and a private note confirming settings restored.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot in Excel · OneDrive

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Sample Prompts — Lab 3

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 3

1

Source controls

Both copies retain120 unique IDs, original owners/statuses and exact cost 35827.50.

2

Display check

Saved S$ two-decimal/date/explanation preferences are checked in two workbooks; observed misses are recorded.

3

Specific request

The output can show35,827.5000 without changing its value or the saved preferences.

4

Restoration

Previous account preferences restored; private baseline note stays outside public courseware.

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Visible .Rules: Instructions Travel with the Workbook

A1	Northstar reporting conventions
A2	Preserve original IDs and current owners
A3	Backlog excludes Closed and Cancelled
A4	Reporting date: 2026-09-30

Formula rule reads Control!B2; Copilot uses the recalculated result on the next request.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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Visible Workbook .Rules and Formula Results

WHAT IT IS

- A visible worksheet named exactly .Rules, including the dot, supplies workbook-specific Copilot instructions in column A.

WHY IT MATTERS

- The workbook can carry stable scope, status, cost-display and preservation instructions alongside the source.

CHECK IT

- The date rule returns Reporting date: 2026-09-30 from Control!B2.

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Visible Workbook .Rules and Formula Results

1. Name the visible sheet exactly .Rules.
2. Place one instruction per column-A cell.
3. Recalculate a formula-driven rule.
4. Check the next output and copied workbook.

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Apply the Concept — Visible Workbook .Rules and Formula Results

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Hidden .Rules is not used; keep the final sheet visible.
- No universal precedence among rules, preferences and Skills is assumed.

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Create and Test Workbook .Rules for Water Service Reporting

LAB 4

Turn Northstar’s workbook conventions into a visible .Rules worksheet. Keep one rule per cell in column A, include a formula-driven reporting-date rule, test the output, and verify that the sheet travels with a copied workbook.

You'll build

Northstar_Rules_Workbook.xlsx with visible .Rules column A, a checked Rules_Test output and a copied-workbook sharing check.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot in Excel · OneDrive

© IES Academy

Sample Prompts — Lab 4

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 4

1

Recognition

Exactly .Rules; visible worksheet; instructions in column A, one idea per cell.

2

Formula rule

Reporting-date formula recalculates and displays2026-09-30; no source values changed.

3

Output controls

120 total;60 backlog;50 overdue;30/48 on-time;S$35,827.50 at All scope.

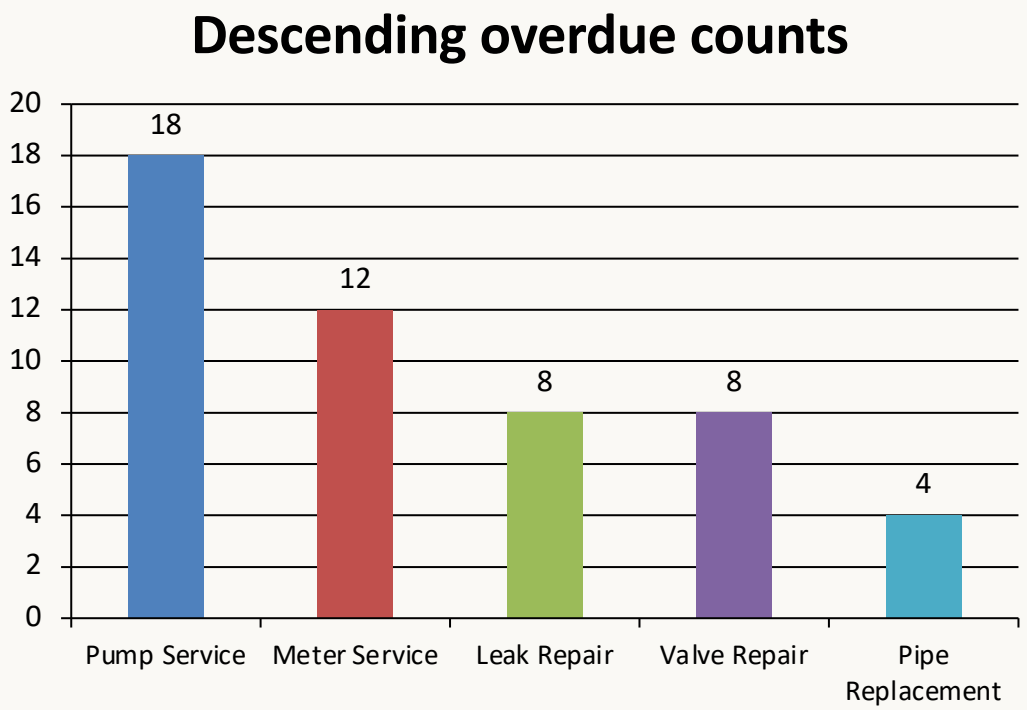
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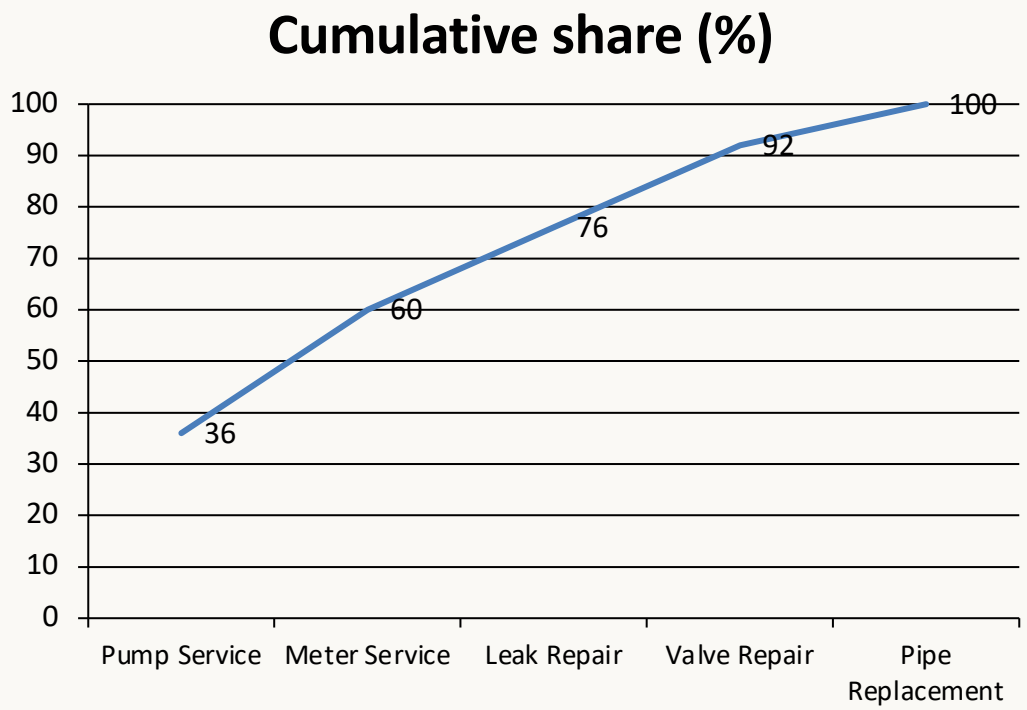
Workbook copy

Visible .Rules travels with the copy; account preferences are not exported.

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Overdue Contribution: 50 Work Orders, Five Categories





Four categories first reach 80%. Equal counts use alphabetical ties. Contribution is not cause.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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Pareto Skill: Counts, Sorting and Cumulative Share

WHAT IT IS

- A Pareto contribution chart orders category counts from largest to smallest and adds a cumulative-percentage line.

WHY IT MATTERS

- The Skill turns a recurring grouping-and-charting method into reusable instructions.

CHECK IT

- Pump 18; Meter 12; Leak 8; Valve 8; Pipe 4: total 50.

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Pareto Skill: Counts, Sorting and Cumulative Share

1

Define source, population, date and grouping.

2

Count and sort with stable ties.

3

Calculate cumulative count / eligible total.

4

Build and verify an editable dual-axis chart.

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Apply the Concept — Pareto Skill: Counts, Sorting and Cumulative Share

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Cumulative percentages are 36%, 60%, 76%, 92%, 100%.
- Four of five categories first reach 80%; do not force an 80/20 conclusion.

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Create a Custom Pareto Chart Skill and Apply It

LAB 5

Author water-service-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue work-order queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping.

You'll build

water-service-pareto/SKILL.md, Northstar_Pareto.xlsx with Pareto_Job_Category and Pareto_Site, and a verified rerun record.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive

© IES Academy

Sample Prompts — Lab 5

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 5

1

Eligible population

Overdue backlog50; no Closed or Cancelled records included.

2

Category counts

Pump Service=18; Meter Service=12; Leak Repair=8; Valve Repair=8; Pipe Replacement=4

3

Cumulative percentages

36%,60%,76%,92%,100%; Leak Repair precedes Valve Repair in the8-count tie.

4

Chart and rerun

Editable columns plus cumulative line; right axis0%-100%; matching output replaced only after review, without duplicates.

5

Source and interpretation

120 original records preserved; Site grouping also totals50; no invented root cause.

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EDITABLE SKILL CONTRACT

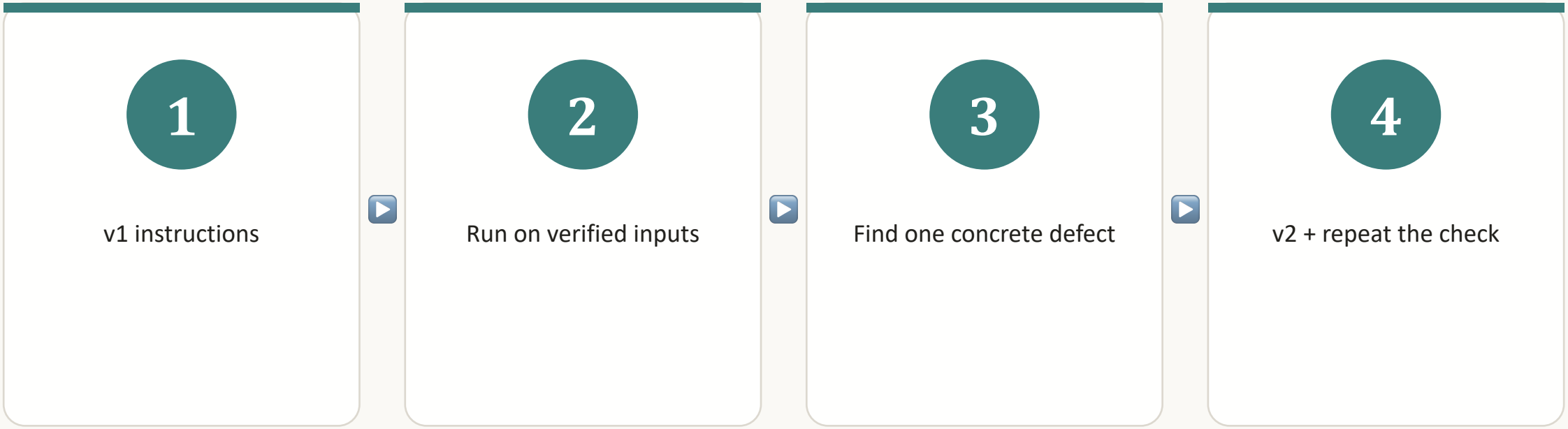
Anatomy of the Maintenance Skill

Discovery	name + description	When should Copilot use it?
Input guard	table or snapshot + date + scope	What evidence is missing?
Method	definitions + source preservation	How should measures be calculated?
Outputs	summary / charts / proposals	Which named areas may change?
Verification	counts + denominators + reruns	What proves the result?

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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A Skill Improves Through Evidence



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Skills, Prompts and Workbook Rules

WHAT IT IS

- A prompt is one request.

WHY IT MATTERS

- The skill makes stable definitions and review boundaries reusable: backlog excludes two terminal statuses, overdue uses a fixed date, missing owners stay missing and output reruns do not duplicate sheets.

CHECK IT

- maintenance-reporting contains definitions and output boundaries.

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Skills, Prompts and Workbook Rules

1. Identify a recurring task.
2. Separate stable rules from current inputs.
3. Write name, description and instructions.
4. Install, invoke and inspect the result.

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Apply the Concept — Skills, Prompts and Workbook Rules

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Each request still supplies this month’s source, date and scope.
- Native skill use and pasted manual instructions are recorded differently.

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Custom Skill Discovery in Each App

WHAT IT IS

- Excel uses its Manage plugins & skills interface and an app-created OneDrive skills folder.

WHY IT MATTERS

- Folder-name/frontmatter mismatches, an extra .txt extension, unavailable licence features and disabled skills can all prevent discovery.

CHECK IT

- Folder maintenance-reporting matches YAML name maintenance-reporting.

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Custom Skill Discovery in Each App

1. Save complete UTF-8 SKILL.md.
2. Use the app-created skills location.
3. Refresh and verify the matching name.
4. Run and record a smoke test.

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Apply the Concept — Custom Skill Discovery in Each App

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Excel and PowerPoint both show the skill before native invocation.
- If a menu is absent, manual practice is labelled feature unavailable.

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Repeatability and Safe Skill Reruns

WHAT IT IS

- Repeatability means the same verified inputs and method yield equivalent business measures and a controlled output structure.

WHY IT MATTERS

- Reporting skills may be used every week or month, so uncontrolled output accumulation quickly creates conflicting summaries.

CHECK IT

- 120 source IDs remain 120 after rerunning Maintenance_Summary.

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Repeatability and Safe Skill Reruns

1. Capture the source fingerprint and outputs.
2. Request a reviewed rerun plan.
3. Replace only named output areas.
4. Compare source and measures again.

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Apply the Concept — Repeatability and Safe Skill Reruns

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Missing report date causes a clarification request, not invented numbers.
- A second output sheet is a defect when replacement was requested.

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Assignment Proposals and Capacity Evidence

WHAT IT IS

- Assignment planning combines the work category, existing owner, estimated hours, technician primary category and available hours.

WHY IT MATTERS

- The same skill can improve Excel analysis and PowerPoint wording by preserving this boundary.

CHECK IT

- Unassigned work is counted only within nonterminal statuses.

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Assignment Proposals and Capacity Evidence

1. Identify unassigned backlog.
2. Compare category and available hours.
3. Record the proposal and assumptions.
4. Request the manager decision.

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Apply the Concept — Assignment Proposals and Capacity Evidence

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Available hours minus proposed effort is a planning check.
- The source Assigned_To stays unchanged until a real authorised operational update.

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Create, Use and Improve the Maintenance Reporting Skill

LAB 6

Reuse the Skill-authoring and installation method from Lab 5 to create maintenance-reporting. Apply it in Excel and PowerPoint, test missing inputs and a controlled rerun, then refine v2 using actual evidence.

You'll build

Northstar_Skill_Report.xlsx with Maintenance_Summary and Assignment_Proposals, Northstar_Skill_Update.pptx, SKILL.md v2 and a before/after improvement log.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel and PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive

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Sample Prompts — Lab 6

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 6

1

Source preserved

120 unique orders; current owner/status unchanged.

2

Skill outputs

Backlog 60; overdue 50; on-time closed 30/48; cost S$35,827.50.

3

Improvement evidence

Input guard and rerun tested; v1/v2 retained; PowerPoint four-slide output accurate.

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Recap - Personalisation, .Rules and Skills for Excel

1. LO2: Personalisation, .Rules, Pareto and reporting Skills tested.

2. Verify: Preserved source plus missing-input and rerun tests.

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TOPIC 3

Interactive Dashboards with Custom Skills

Create the Skill, build the dashboard and prove interactions

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Key Concepts — Interactive Dashboards with Custom Skills

1

Measures

Counts, backlog, overdue, on-time rate and cost.

2

Visuals

Choose charts by maintenance question.

3

Interactions

One filter scope for every chart and KPI.

4

Refresh

Known deltas in an isolated test copy.

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Hands-On Labs — Interactive Dashboards with Custom Skills

Lab 7

- **Custom Dashboard Skill**
  - Create and Apply a Custom Interactive Dashboard Skill

Lab 8

- **Interactive Filters and Refresh**
  - Use the Dashboard Skill for Filters, Refresh and Safe Reruns

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WORKED DASHBOARD SNAPSHOT

Northstar Portfolio at a Glance

Total orders

120

Backlog

60

Overdue

50

Closed

48

On-time closed

30/48

Recorded cost

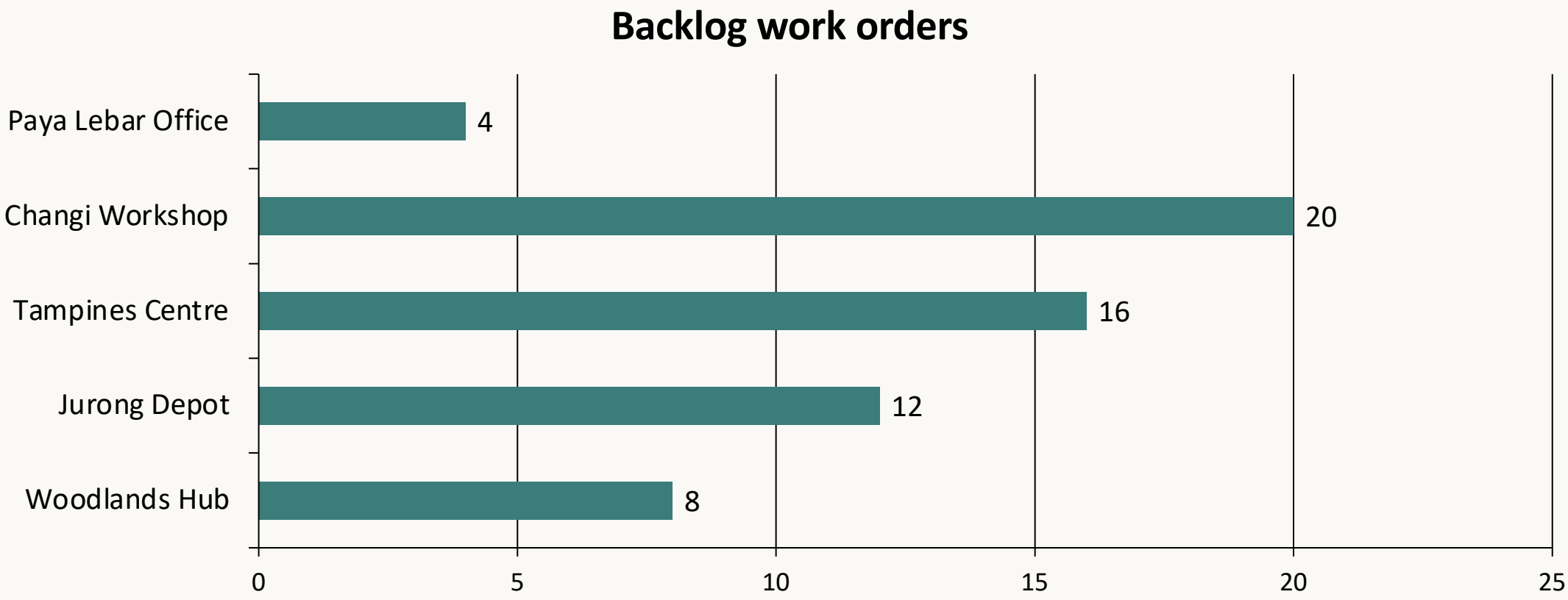
S$35,828

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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NATIVE EDITABLE CHART

Backlog by Site: Count the Nonterminal Work



Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

The Interactive Dashboard Skill Contract

1. Named source + date
2. Six shared-scope KPIs
3. Four filters + three charts
4. Refresh + safe rerun checks

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Custom Dashboard Skill as a Reusable Contract

WHAT IT IS

- water-service-dashboard is a separate custom Skill whose instructions define inputs, six KPI formulas, four shared filter selectors, three editable charts and controlled output boundaries.

WHY IT MATTERS

- A reusable contract makes every KPI and chart obey the same AND filter logic and allows a safe refresh using table references.

CHECK IT

- Dashboard B2:B5 select site, category, status and owner.

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Custom Dashboard Skill as a Reusable Contract

1. Author and review the complete SKILL.md.
2. Install and verify the dashboard name.
3. Invoke a plan and build the reviewed output.
4. Test common filters, append refresh and reruns.

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Apply the Concept — Custom Dashboard Skill as a Reusable Contract

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- B7:B12 and all chart summaries use the identical selected rows.
- A 122-row sandbox is separate from the untouched 120-row baseline.

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Dashboard Measures and Denominators

WHAT IT IS

- A dashboard measure needs a business definition, numerator, denominator where relevant, date rule and filter scope.

WHY IT MATTERS

- Changing the denominator can make an attractive rate meaningless.

CHECK IT

- Closed-only rate is different from closed/total completion share.

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Dashboard Measures and Denominators

1. Define source fields and scope.
2. Compute count and denominator.
3. Handle empty selections explicitly.
4. Reconcile against controls.

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Apply the Concept — Dashboard Measures and Denominators

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A due-today backlog item is not overdue under a strict before-date rule.
- All selected status/chart counts reconcile to total selected work orders.

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Visual Dashboard Design for Maintenance

WHAT IT IS

- A dashboard should make an operational decision easier.

WHY IT MATTERS

- Consistent chart scales, short labels, whitespace and limited accents help a manager compare the work queue quickly.

CHECK IT

- Status count chart shows the portfolio mix.

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Visual Dashboard Design for Maintenance

1. Name the maintenance decision.
2. Choose the measure and comparison.
3. Use an appropriate labelled chart.
4. Check meaning and readability.

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Apply the Concept — Visual Dashboard Design for Maintenance

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Site backlog chart excludes Closed and Cancelled.
- A category count does not establish the root cause of overdue jobs.

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Create and Apply a Custom Interactive Dashboard Skill

LAB 7

Create water-service-dashboard/SKILL.md for the Northstar maintenance workflow, install it in Excel, then invoke it to plan and build an interactive dashboard. Six KPIs and three charts share four filter selections.

You'll build

Northstar_Dashboard.xlsx with a Dashboard layout, six verified measures, three labelled charts and a measure dictionary ready for Lab 8 filters.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive

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Sample Prompts — Lab 7

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 7

1

Baseline

Total 120; backlog 60; overdue 50; closed 48.

2

SLA and cost

30/48 = 62.5%; S$35,827.50.

3

Chart reconciliation

Three count summaries each total 120; formulas reference tblWorkOrders.

4

Decision design

Measures dictionary and source/date visible; no unsupported cause claims.

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EDITABLE WORKED COMPARISON

Same Data, Different Filter Scope

All sites

Total: 120

Backlog: 60

Overdue: 50

Closed: 48

On-time closed: 30/48

Woodlands Hub

Total: 24

Backlog: 8

Overdue: 7

Closed: 16

On-time closed: 10/16

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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One Filter Scope for Every Visual

WHAT IT IS

- Interactive means a learner-controlled selection changes the evidence shown.

WHY IT MATTERS

- A dashboard can appear interactive while only one chart changes.

CHECK IT

- Woodlands Hub plus All in other selectors isolates that site.

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One Filter Scope for Every Visual

1. Define the selector cells.
2. Apply the same AND mask everywhere.
3. Test each control and edge case.
4. Reconcile selected chart and KPI totals.

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Apply the Concept — One Filter Scope for Every Visual

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Status New gives zero Closed and N/A closure rate.
- A slicer must connect to every intended PivotTable.

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Refresh Tests and Static Snapshots

WHAT IT IS

- A refresh test deliberately introduces known new records and verifies the expected delta.

WHY IT MATTERS

- A correct dashboard today may fail after the next export if formulas reference a fixed cell range.

CHECK IT

- Two test rows increase total by2 and recorded cost byS$400.

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Refresh Tests and Static Snapshots

1. Save an isolated sandbox copy.
2. Append known unique test rows.
3. Recalculate and verify expected deltas.
4. Handoff the untouched baseline.

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Apply the Concept — Refresh Tests and Static Snapshots

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Only one new row adds to backlog/overdue; the other adds to on-time Closed.
- The final deck uses the untouched 120-row baseline, not the test copy.

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Use the Dashboard Skill for Filters, Refresh and Safe Reruns

LAB 8

Continue Lab 7 by making four dropdown selectors control every KPI and chart. Test a single site, a zero-denominator selection and a combined category/owner filter.

You'll build

Northstar_Dashboard.xlsx with four working filters plus Northstar_Dashboard_Refresh_Test.xlsx, filter evidence and a verified baseline snapshot for PowerPoint.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive

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Sample Prompts — Lab 8

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 8

1

Woodlands control

Total 24; backlog 8; overdue 7; closed 16; cost S$6,668.50; on-time 10/16.

2

Interactive scope

Changing any of four filters affects all six KPI measures and all chart counts.

3

Edge case

No Closed -> N/A; no rows -> zero count/cost and N/A rate.

4

Refresh sandbox

122 total;49 closed;61 backlog;51 overdue;S$36,227.50;31/49 on-time.

5

Handoff baseline

Return to 120-row untouched baseline; date and All scope recorded.

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Recap - Interactive Dashboards with Custom Skills

1. LO3: Six measures and three charts share four filters.

2. Verify: Baseline controls and known refresh deltas.

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TOPIC 4

Create a Professional Presentation with PowerPoint Copilot

Continue the dashboard into a decision pack

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Key Concepts — Create a Professional Presentation with PowerPoint Copilot

1

Evidence story

Observation, implication, action and decision.

2

Charts

Reviewed values, provenance and snapshot date.

3

Refinement

Hierarchy, notes and actual rehearsal feedback.

4

Handoff

Reconciled pack and unsent communication.

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Hands-On Labs — Create a Professional Presentation with PowerPoint Copilot

Lab 9

- **Dashboard to presentation**
  - Turn Dashboard Evidence into a Management Story

Lab 10

- **Professional handoff**
  - Refine and Present the Maintenance Decision Deck

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A Verified Chart Becomes a Decision

1. Observed backlog
2. Operational implication
3. Proposed recovery action
4. Manager decision requested

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From Dashboard Evidence to a Decision Story

WHAT IT IS

- A management presentation transforms evidence into a sequence of observations, implications, proposed actions and a decision request.

WHY IT MATTERS

- Copilot can accelerate outlines and drafting, but a persuasive invented conclusion is still wrong.

CHECK IT

- Backlog by site supports prioritisation, not a proven cause.

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From Dashboard Evidence to a Decision Story

1. Freeze the verified dashboard snapshot.
2. Register each proposed claim.
3. Write a reviewed Word brief.
4. Generate and reconcile the deck.

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Apply the Concept — From Dashboard Evidence to a Decision Story

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Available-hours evidence supports a capacity discussion, not confirmed allocation.
- The final slide asks management to decide on a proposed plan.

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Charts, Templates and Source Provenance

WHAT IT IS

- A template supplies slide layouts, typography and colours; it does not supply correct facts.

WHY IT MATTERS

- Approximate AI-generated charts can contradict source values while looking plausible.

CHECK IT

- An editable embedded chart remains a copy rather than a tested live link.

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Charts, Templates and Source Provenance

1. Select the reviewed brief.
2. Apply the existing template.
3. Insert verified charts or tables.
4. Record source and snapshot scope.

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Apply the Concept — Charts, Templates and Source Provenance

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- An image is labelled as a snapshot at 30 September.
- All eight slides use one reviewed baseline.

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Turn Dashboard Evidence into a Management Story

LAB 9

Continue the baseline dashboard from Topic 3. Build a slide evidence register and a Word presentation brief before asking PowerPoint Copilot for an eight-slide management deck.

You'll build

Northstar_Presentation_Brief.docx, Northstar_Management_Draft.pptx and a slide-by-slide evidence register tied to the 120-row dashboard snapshot.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive

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Sample Prompts — Lab 9

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 9

1

Evidence continuity

Every deck figure maps to the 120-row Lab 8 baseline and slide evidence register.

2

Eight-slide narrative

Observation -> implication -> proposed action -> decision; no invented cause/savings.

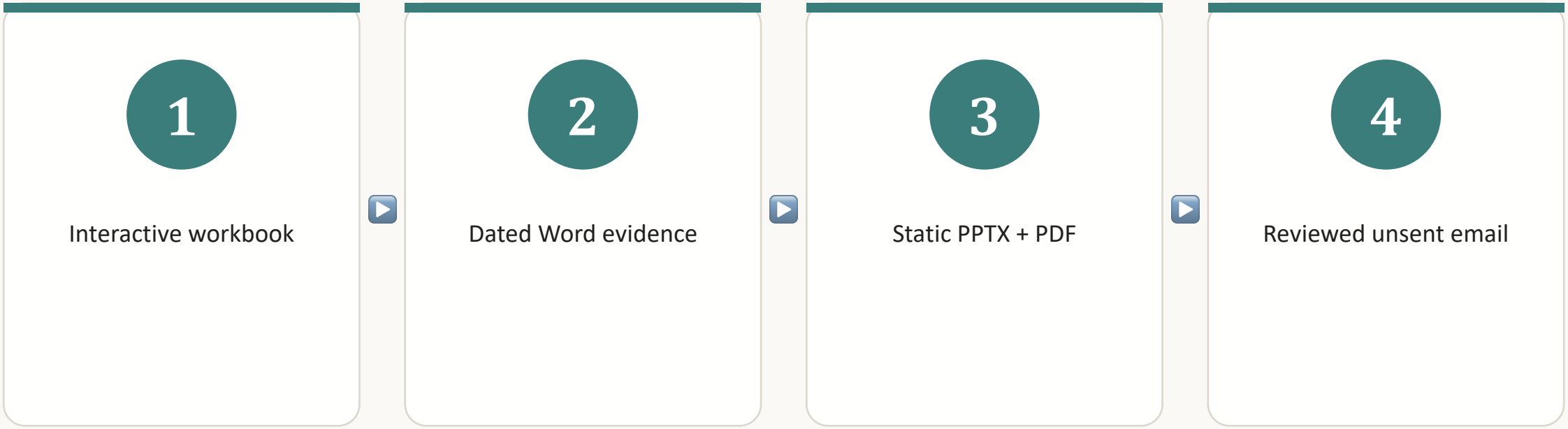
3

Chart semantics

Backlog charts exclude Closed/Cancelled; embedded/image copies labelled snapshot.

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The Maintenance Reporting Handoff



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Professional Refinement and Rehearsal

WHAT IT IS

- Refinement improves the clarity of the existing evidence without changing its meaning.

WHY IT MATTERS

- A well-designed deck can still fail when the speaker’s notes contain different numbers or a proposal is spoken as an approved action.

CHECK IT

- One message per slide; chart labels remain readable.

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Professional Refinement and Rehearsal

1. Audit the draft against evidence.
2. Apply reviewed design changes.
3. Rehearse and record actual feedback.
4. Inspect PPTX and exported PDF.

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Apply the Concept — Professional Refinement and Rehearsal

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A 45-second note explains evidence and limitation.
- Partner feedback leads to one recorded revision.

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Reviewed Handoff and Artifact Lifecycle

WHAT IT IS

- The final pack contains an interactive Excel dashboard, a fixed reporting brief, a static PowerPoint/PDF snapshot, the custom skill and a reviewed communication draft.

WHY IT MATTERS

- Updating one artifact without the others creates inconsistent communication.

CHECK IT

- PowerPoint and PDF both show eight slides/pages from the same baseline.

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Reviewed Handoff and Artifact Lifecycle

1. Reconcile final artifacts.
2. Record actual filenames and links.
3. Review the handoff draft.
4. Preserve the dated snapshot and skill.

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Apply the Concept — Reviewed Handoff and Artifact Lifecycle

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Outlook does not claim a missing attachment was sent.
- The next update reruns the method with new verified inputs.

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Refine and Present the Maintenance Decision Deck

LAB 10

Refine the management deck using the same custom skill and reviewed dashboard evidence. Improve hierarchy, consistency and speaker notes, test the story in a short rehearsal, and prepare a reviewed Outlook handoff draft.

You'll build

Northstar_Management_Final.pptx/PDF, a rehearsal record, final evidence register and unsent Outlook handoff draft with accurate artifact links.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive

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Sample Prompts — Lab 10

1. Scope: State actual source, report date and selected scope before acting.

2. Verify: Explain the source/formula checks and preserve unknowns and original records.

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Check Your Results — Lab 10

1

Presentation integrity

Eight slides and eight PDF pages; visible figures agree with verified register.

2

Professional delivery

Readable chart labels, consistent design, evidence-specific notes and actual rehearsal improvement.

3

Final continuity

Baseline dashboard, brief, deck and unsent draft agree; static snapshot stated.

4

Decision boundary

No real email sent or work order reassigned; proposed actions remain proposed.

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Recap - Create a Professional Presentation with PowerPoint Copilot

1. LO4: Eight-slide evidence story and reviewed handoff.

2. Verify: PDF, notes, register and actual rehearsal agree.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Connected Copilot workflow

Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.

2

Reusable maintenance skills

Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.

3

Interactive dashboard

Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.

4

Professional presentation

Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.

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NEXT STEPS

Your Next Maintenance Report

1. Use a copy and a new explicit reporting date.
2. Supply verified source/date/scope and test skill reruns.
3. Reconcile every claim and decision boundary.

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Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded IES Academy course files and use the lab checkpoints to restart safely.
4. Keep practising with the IES Academy course materials.

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THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.

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